Leader Selection and Coherence Optimization in Sparse Power-Network Graphs: A Five-Dataset Graph-Surrogate Validation*

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Abstract

Network coherence measures the steady-state variance of a networked dynamical system under persistent stochastic disturbance. This paper studies coherence reduction in sparse power-network graphs through grounded-Laplacian leader selection, learning-assisted candidate screening, and bounded edge-weight reallocation. Five MatrixMarket power-network datasets are analyzed: HB/662_bus, HB/685_bus, HB/1138_bus, HB/eris1176, and Pajek/USpowerGrid. The three HB bus admittance matrices are converted into weighted graphs using mean-normalized off-diagonal admittance magnitudes. The eris1176 and US-grid files are pattern matrices and are therefore treated as unweighted topologies; because eris1176 contains a separate two-node component, experiments are conducted on its 1174-node largest connected component. We formulate first-order stochastic leader-follower dynamics, define normalized coherence as the scaled trace of the inverse grounded weighted Laplacian, and implement an exact greedy method using a Laplacian pseudoinverse identity and Schur-complement updates. A random-forest screened greedy method learns marginal leader utilities from structural features and evaluates only a screened subset of candidates exactly. Across the five validation graphs, exact greedy leader selection reduces $k = 20$ coherence relative to random leaders by 17.49%, 20.86%, 21.92%, 30.94%, and 17.67%, respectively. The screened method evaluates approximately one quarter of candidates per step and remains within -0.49% to +0.15% of the exact greedy trajectory at $k = 20$. For fixed ten-leader sets on weighted admittance graphs, bounded edge-weight reallocation gives further reductions of 32.75% on 662-bus and 33.33% on 1138-bus, with a 14.69% reduction in a two-step diagnostic run on the newly added 685-bus graph. These results support grounded-Laplacian coherence optimization as an interpretable graph-level robustness tool for sparse infrastructure networks, while clarifying that engineering deployment requires additional power-flow, operating-point, and contingency validation.

Keywords: network coherence; leader-follower consensus; grounded Laplacian; power-grid graph; leader selection; random forest; edge-weight allocation; complex networks.

1 Introduction

Power systems, communication networks, transportation systems, and multi-agent platforms are all networked infrastructures whose global behavior depends on how local disturbances propagate through sparse coupling graphs. In stochastic consensus models, random perturbations at nodes

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are filtered by a graph Laplacian. The resulting steady-state variance, usually called *network coherence*, is a compact performance measure: smaller coherence indicates better attenuation of persistent noise and stronger anchoring of weak Laplacian modes [1, 2].

This paper considers the design problem in which a limited number of nodes are selected as leaders, references, or pinned nodes. In a power-network graph, these leaders should be interpreted as graph-theoretic monitoring or control anchors in a linear stochastic surrogate, not as direct recommendations for generator placement, protection settings, or physical line upgrades. The mathematical question is nevertheless important: which nodes most reduce the trace of the inverse grounded Laplacian, and can this spectral objective be accelerated by statistical learning?

The main improvement over a single-network study is the use of five validation datasets with different sizes, weights, and structural bottlenecks. The newly added 685-bus admittance graph provides a third weighted HB bus matrix. The newly added eris1176 file is especially useful as a stress test because it is not an admittance-weighted matrix; it is a symmetric pattern matrix with a large connected component, high clustering, many articulation points, and a small disconnected two-node component. Its inclusion forces the paper to state precisely how disconnected pattern graphs are handled under grounded-Laplacian coherence.

The contributions are as follows. First, a transparent data-processing pipeline is given for MatrixMarket power-network files, including weighted-admittance construction, pattern-graph construction, and largest-connected-component handling for disconnected data. Second, exact greedy leader selection is implemented using a pseudoinverse identity for the first leader and exact Schur-complement updates for all later leaders. Third, a random-forest screened greedy method is evaluated across all five graphs and shown to preserve almost all spectral-greedy performance while reducing exact candidate evaluations to about one quarter. Fourth, bounded edge-weight reallocation is revisited on the weighted admittance graphs to quantify how much additional coherence can be gained after leader placement.

The manuscript is deliberately positioned as a graph-surrogate study. It does not claim that the selected leaders are immediately deployable control devices or that the reallocated weights are feasible line upgrades. This positioning is important for review: the contribution is an interpretable spectral-design workflow and multi-dataset validation, whereas engineering deployment would require power-flow and contingency studies beyond the information contained in the MatrixMarket files.

2 Data and graph construction

2.1 MatrixMarket archives

The analysis uses five MatrixMarket power-network archives downloaded from the Network Repository power-network page [9]: `https://networkrepository.com/power.php`. The files used in the experiments are:

- `power-662-bus.zip`
- `power-685-bus.zip`
- `power-1138-bus.zip`
- `power-eris1176.zip`
- `power-US-Grid.zip`

The first three files are real symmetric bus admittance matrices from the HB collection. The last two are pattern symmetric matrices and do not contain physical admittance magnitudes. All node labels in the result tables are reported using one-indexed labels.

For an HB admittance matrix A, each nonzero off-diagonal entry defines an undirected edge. Since the off-diagonal entries are negative couplings in the uploaded admittance matrices, the raw edge-coupling magnitude is taken as

$$c_{ij} = |A_{ij}|, \qquad i \neq j. \tag{1}$$

The graph weight is mean-normalized,

$$w_{ij} = \frac{c_{ij}}{|E|^{-1} \sum_{(u,v)\in E} c_{uv}}, \tag{2}$$

so the average edge weight is one. This normalization does not change relative admittance magnitudes; it only fixes the unit scale of coherence. Diagonal entries of the original admittance matrices are not reused as graph-Laplacian diagonals, because they may include shunt or modeling terms. Throughout the paper, an edge weight means this normalized off-diagonal magnitude; it should not be read as a directly dispatchable physical conductance or a permitted line-investment decision. For pattern matrices, every edge receives unit weight.

The eris1176 pattern graph has two connected components of sizes 1174 and 2. A standard grounded-Laplacian inverse is positive definite only if every connected component contains at least one leader. To keep the same single-component experimental protocol used for all other graphs, the reported eris1176 results use the 1174-node largest connected component (LCC). For the full disconnected graph, a component-aware protocol would first ensure that every component receives at least one leader, or would minimize a sum of component-level coherence terms under an explicit leader-budget allocation. This extension is not pursued here because the isolated two-node component would otherwise dominate a small part of the feasibility logic rather than the main spectral-placement question.

2.2 Network statistics

Table 1 summarizes the parsed graphs. The data are all sparse, but they differ markedly. The 685-bus graph has more edges than 662-bus, fewer bridges than both 662-bus and 1138-bus, and higher clustering than the other weighted admittance graphs. The eris1176-LCC graph has many more edges and a high maximum degree, yet it still contains 309 bridges and 255 articulation points. The US grid is the largest graph and contains 1611 bridges. These bottlenecks are precisely the structures where purely local centrality heuristics can fail to identify the most useful leaders.

Table 1: Network statistics after graph construction. HB edge weights are mean-normalized admittance magnitudes; pattern graphs use unit weights. The eris1176 row reports its largest connected component.

Dataset	Type	N	$\|E\|$	Density	Avg. deg.	Max deg.	Art.	Bridges	Diam.	Avg. dist.	Med. w	Max w
662-bus	weighted admittance	662	906	0.004141	2.737	9	80	88	25	10.24	0.3372	49.63
685-bus	weighted admittance	685	1282	0.005472	3.743	12	34	31	26	12.42	0.4466	137.99
1138-bus	weighted admittance	1138	1458	0.002254	2.562	17	361	473	31	12.72	0.0608	29.99
eris1176-LCC	unweighted pattern	1174	8687	0.012616	14.799	98	255	309	53	12.06	1.0000	1.00
US Grid	unweighted pattern	4941	6594	0.000540	2.669	19	1229	1611	46	18.99	1.0000	1.00

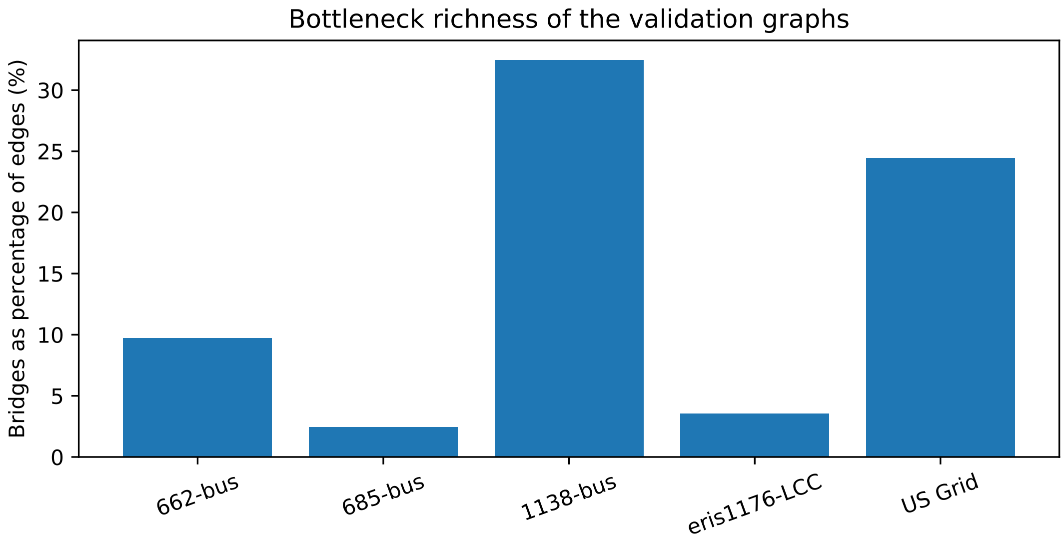


Figure 1: Fraction of edges that are bridges. Bridge-rich networks are challenging for coherence design because a poorly placed leader can leave entire corridors weakly anchored.

3 Leader-follower coherence model

Let $G = (V, E, w)$ be a connected weighted graph with $N = |V|$. The weighted Laplacian is

$$L(w) = \sum_{(i,j)\in E} w_{ij}(e_i - e_j)(e_i - e_j)^\top, \tag{3}$$

where e_i is the ith standard basis vector. For a nonempty leader set $S \subset V$, let $F = V \setminus S$ denote the follower set. After ordering followers first, the Laplacian is partitioned as

$$L(w) = \begin{bmatrix} L_{FF}(S, w) & L_{FS}(S, w) \\ L_{SF}(S, w) & L_{SS}(S, w) \end{bmatrix}. \tag{4}$$

If G is connected and S is nonempty, the grounded Laplacian $L_{FF}(S, w)$ is symmetric positive definite. The stochastic follower dynamics are

$$dx_F(t) = -L_{FF}(S, w)x_F(t)\,dt + dB_t, \tag{5}$$

where B_t is standard Brownian motion. The steady-state covariance P_S satisfies

$$L_{FF}P_S + P_S L_{FF} = I, \tag{6}$$

so $P_S = \frac{1}{2}L_{FF}^{-1}$ for symmetric L_{FF}. The normalized coherence objective is

$$H(S, w) = \frac{1}{2N}\operatorname{tr}\left(L_{FF}(S, w)^{-1}\right). \tag{7}$$

Smaller H means smaller aggregate steady-state variance around the leader-pinned reference.

The joint design problem is

$$\begin{aligned} \min_{S,w} \quad & H(S, w) \\ \text{s.t.} \quad & |S| = k, \quad S \subset V, \\ & \sum_{e\in E} w_e = B, \quad \ell_e \le w_e \le u_e. \end{aligned} \tag{8}$$

The leader-selection part is combinatorial. For fixed S, the edge-weight subproblem is convex because $L_{FF}(S,w)$ is affine in w and $X \mapsto \mathrm{tr}(X^{-1})$ is convex on the positive definite cone [11, 12]. Equation (8) is therefore best understood as the full design template. The experiments below solve it sequentially: leader sets are selected first, and bounded edge-weight reallocation is then tested for fixed leaders on the weighted admittance graphs. No global optimality claim is made for the combined mixed discrete–continuous problem.

4 Algorithms

4.1 Exact greedy leader selection

The exact greedy rule selects one leader at a time. Given a current leader set S, the next leader is

$$p^\star = \arg\max_{p \in F} \left\{ \mathrm{tr}\left(L_{FF}(S,w)^{-1}\right) - \mathrm{tr}\left(L_{F\setminus\{p\},F\setminus\{p\}}(S \cup \{p\}, w)^{-1}\right) \right\}. \tag{9}$$

The first leader can be selected from the Moore-Penrose inverse L^+ of the full Laplacian. If node r is grounded, then

$$\mathrm{tr}\left(L_{\bar{r},\bar{r}}^{-1}\right) = N(L^+)_{rr} + \mathrm{tr}(L^+), \tag{10}$$

so the best single leader is the node with the smallest diagonal entry of L^+.

For later steps, suppose $X = L_{FF}(S,w)^{-1}$ and candidate p is the first follower after reordering:

$$X = \begin{bmatrix} \alpha & \beta^\top \\ \beta & \Gamma \end{bmatrix}. \tag{11}$$

After grounding p, the new inverse is

$$X_{-p} = \Gamma - \frac{\beta\beta^\top}{\alpha}. \tag{12}$$

The exact marginal trace reduction is therefore

$$\Delta(p \mid S) = \mathrm{tr}(X) - \mathrm{tr}(X_{-p}) = \alpha + \frac{\|\beta\|_2^2}{\alpha} = \frac{\|X_{p,:}\|_2^2}{X_{pp}}. \tag{13}$$

This identity gives all current candidate gains from one inverse matrix and updates the inverse exactly after each greedy choice.

4.2 Computational cost and scaling

The implementation used for this validation maintains a dense inverse of the current grounded Laplacian. After the first inverse is formed, each greedy step can compute all marginal gains from rows of the inverse and then apply a rank-one Schur-complement update. This is efficient for the five benchmark graphs considered here, but it still requires $O(N^2)$ memory and dense linear-algebra operations. For much larger transmission or distribution networks, the same objective would need sparse selected-inversion, iterative trace estimation, or localized candidate generation. The random-forest screen reduces the number of exact candidate-gain evaluations, but it does not by itself remove the need to update the grounded inverse along the selected trajectory.

4.3 Centrality baselines

The baselines are degree, weighted degree, core number, closeness, harmonic centrality, and betweenness. For weighted admittance graphs, path-based centralities use distance $d_{ij} = 1/w_{ij}$ so that stronger admittance-magnitude coupling corresponds to shorter graph distance. For unweighted pattern graphs, all distances are unit distances. For the US grid, harmonic and betweenness rankings are sampled approximations used only for ranking; the reported coherence values are exact for the selected leader sets.

4.4 Random-forest screened greedy

The random-forest screened method uses exact spectral gains as labels but avoids evaluating every candidate exactly at every step. At step t, a candidate node receives static graph features—degree, weighted degree, clustering, closeness, harmonic centrality, betweenness, articulation indicator, and core number—plus dynamic distances to the current leader set. A random subset of candidates is evaluated exactly using Eq. (13). A random forest regressor is trained on these labels. The algorithm then exactly evaluates the training subset plus the top predicted candidates and selects the node with the largest exact gain among that screened set.

The default screen evaluates 15% of candidates at random and 12% by predicted rank, with a minimum of 80 candidates in each part. The final choice at each step is always based on exact grounded-Laplacian gain, not on the regressor alone. Thus the learning model is used only for candidate prioritization; it is not used to report approximate objective values. For eris1176-LCC, the regressor uses an approximate betweenness feature to avoid excessive preprocessing time; all selected-set coherence values remain exact.

4.5 Bounded edge-weight reallocation

For fixed leaders, let $b_{e,F}$ be the reduced incidence vector of edge e after deleting leader coordinates. Then

$$L_{FF}(S, w) = \sum_{e \in E} w_e b_{e,F} b_{e,F}^\top. \tag{14}$$

The gradient of Eq. (7) is

$$\frac{\partial H(S, w)}{\partial w_e} = -\frac{1}{2N} b_{e,F}^\top L_{FF}^{-2} b_{e,F} = -\frac{1}{2N} \| L_{FF}^{-1} b_{e,F} \|_2^2. \tag{15}$$

The weighted-edge experiment keeps the total normalized edge weight fixed and constrains every edge to move by at most 50%:

$$\sum_{e \in E} w_e = \sum_{e \in E} w_e^{(0)}, \qquad 0.5 w_e^{(0)} \le w_e \le 1.5 w_e^{(0)}. \tag{16}$$

Projection onto this box-simplex set is computed by bisection on a scalar Lagrange multiplier.

5 Experimental protocol

All reported node IDs are one-indexed. The random seed is fixed at 20260525. Exact greedy leader selection is run to $k = 20$. Random baselines are means over 12 random leader orderings with one-standard-deviation bands; these bands are used as a diagnostic variability summary rather than as formal confidence intervals. The random-forest screening procedure is also run to $k = 20$ for every validation graph. Weighted-edge reallocation is applied only to the three weighted admittance graphs because the pattern graphs do not contain observed admittance-magnitude coupling weights. For the 662-bus and 1138-bus graphs, the weight-reallocation runs use 20 projected-gradient iterations. For the newly added 685-bus graph, Table 6 reports a conservative two-step diagnostic run; it should be interpreted as a lower-bound demonstration of additional improvement rather than as a fully converged edge-design result.

6 Results

6.1 Leader selection across five datasets

Table 2 reports coherence values at $k = 10$ and $k = 20$. Exact greedy consistently gives the best or near-best value among the reported deterministic selection rules. At $k = 20$, the reduction relative to random leaders is 17.49% on 662-bus, 20.86% on 685-bus, 21.92% on 1138-bus, 30.94% on eris1176-LCC, and 17.67% on the US grid.

Table 2: Coherence comparison for selected leader counts. Smaller H is better. The last column compares exact greedy with the random mean.

Dataset	k	Greedy	RF	Degree	W-degree	Betweenness	Harmonic	Random mean	Greedy vs random
662-bus	10	2.4992	2.4997	2.8727	2.9243	2.9952	3.0475	2.8722 ± 0.0530	12.99%
662-bus	20	2.1791	2.1788	2.7009	2.8779	2.9058	3.0138	2.6411 ± 0.0756	17.49%
685-bus	10	1.1493	1.1521	1.4784	1.5957	1.5951	1.8952	1.4443 ± 0.0603	20.43%
685-bus	20	0.9808	0.9761	1.1468	1.3436	1.5164	1.8093	1.2394 ± 0.0532	20.86%
1138-bus	10	24.7609	24.7936	29.0617	28.6054	29.6824	30.4835	29.3737 ± 0.7039	15.70%
1138-bus	20	21.2177	21.2504	26.3199	27.6721	29.1543	29.3179	27.1737 ± 0.5013	21.92%
eris1176-LCC	10	0.6107	0.6111	0.9760	0.9760	0.8985	0.9120	0.8302 ± 0.0356	26.44%
eris1176-LCC	20	0.4964	0.4971	0.9759	0.9759	0.8799	0.9116	0.7188 ± 0.0616	30.94%
US Grid	10	1.0058	1.0024	1.1245	1.1245	1.2174	1.3229	1.2445 ± 0.0342	19.18%
US Grid	20	0.9141	0.9096	0.9991	0.9991	1.1520	1.2295	1.1103 ± 0.0221	17.67%

Figure 2 normalizes the $k = 20$ coherence values by the random mean for each graph. This relative view makes the cross-dataset pattern visible despite the different absolute coherence scales. Greedy and RF-screened greedy are consistently lower than random and centrality-based choices. The eris1176-LCC case is notable: it has high average degree, but centrality baselines remain far from spectral greedy because many high-degree regions are redundant while several weakly anchored corridors dominate the trace-of-inverse objective.

Table 3 compares exact greedy with the best centrality baseline at $k = 20$. Even the best centrality is 9.30% worse than greedy on the US grid and 77.26% worse on eris1176-LCC. This confirms that grounded-Laplacian coherence is not reducible to a purely local or shortest-path centrality score.

Table 3: Best centrality baseline at $k = 20$ and its gap relative to exact greedy.

Dataset	Best centrality	$H_{\text{centrality}}$	Gap vs greedy	Greedy vs random
662-bus	Degree	2.7009	23.95%	17.49%
685-bus	Degree	1.1468	16.92%	20.86%
1138-bus	Degree	26.3199	24.05%	21.92%
eris1176-LCC	Betweenness	0.8799	77.26%	30.94%
US Grid	Degree	0.9991	9.30%	17.67%

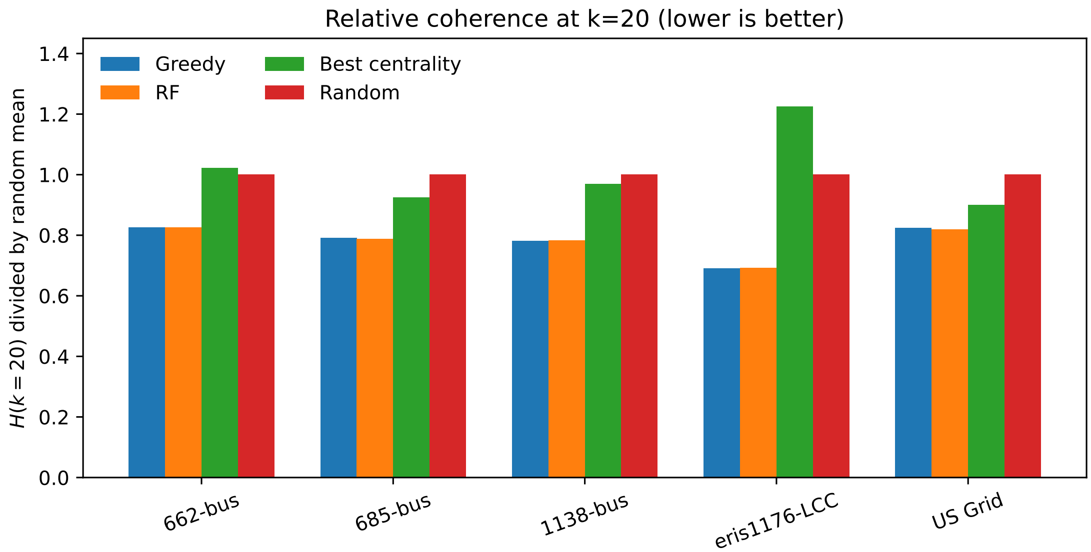


Figure 2: Relative coherence at $k = 20$, normalized by the random-leader mean for each dataset. Lower is better. "Best centrality" is the best among degree, weighted degree, betweenness, and harmonic centrality for that dataset.

Table 4: First ten exact greedy leaders. IDs are one-indexed labels from the corresponding Matrix-Market graph or largest connected component mapping.

Dataset	First ten greedy leaders
662-bus	393, 201, 502, 45, 639, 230, 417, 282, 294, 272
685-bus	180, 341, 295, 237, 487, 100, 553, 376, 545, 511
1138-bus	536, 877, 395, 861, 1089, 241, 856, 772, 608, 100
eris1176-LCC	752, 982, 379, 501, 1127, 1003, 936, 526, 495, 80
US Grid	1243, 4164, 3312, 316, 2554, 726, 4345, 1166, 4652, 1005

6.2 Learning-assisted screening

Table 5 summarizes random-forest screening. The method evaluates about 25% of candidates per step. The $k = 20$ gap is small in every case, ranging from -0.485% on the US grid to +0.154% on the 1138-bus graph. A negative gap does not imply global superiority of the screening method; it means that exact greedy is a myopic trajectory and the screened path can occasionally make a different early choice that gives slightly lower coherence at a fixed later cardinality.

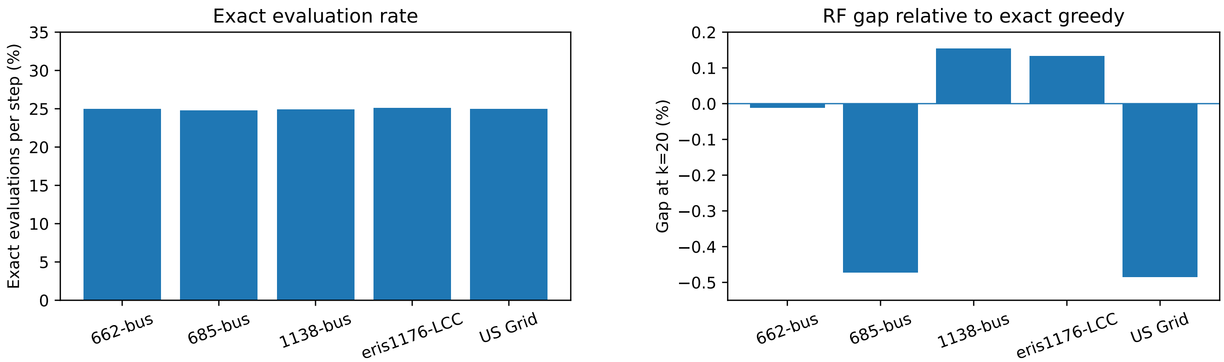
6.3 Effect of the newly added datasets

The 685-bus graph strengthens the weighted-admittance part of the study. It has more edges than the 662-bus graph, fewer bridges than both older weighted cases, and higher clustering than the other weighted admittance graphs; nevertheless, exact greedy still reduces $k = 20$ coherence by 20.86% relative to random leaders and is 16.92% lower than the best centrality baseline. This suggests that the spectral leader-selection advantage is not limited to extremely bridge-dominated weighted graphs.

The eris1176-LCC graph strengthens the pattern-topology part of the study. It is denser than

Table 5: Random-forest screened greedy diagnostics. Correlations and R^2 are averages over greedy steps 2 through 20. The $k = 20$ gap is $(H_{\mathrm{RF}} - H_{\mathrm{greedy}})/H_{\mathrm{greedy}}$.

Dataset	Eval. pct.	Avg. eval.	Exact-hit	Pearson	Spearman	R^2	$k = 20$ gap
662-bus	24.99	162.9	18/19	0.779	0.784	0.592	-0.012%
685-bus	24.76	167.2	13/19	0.768	0.785	0.585	-0.473%
1138-bus	24.91	281.0	17/19	0.797	0.805	0.626	+0.154%
eris1176-LCC	25.09	292.0	15/19	0.907	0.916	0.820	+0.134%
US Grid	24.95	1230.3	15/19	0.815	0.836	0.659	-0.485%



(a) Exact evaluations.

(b) $k = 20$ gap.

Figure 3: Screened greedy reduces the exact candidate-evaluation burden while keeping the final coherence close to exact greedy.

the other validation graphs, with average degree 14.80 and maximum degree 98, but its $k = 20$ centrality gap is the largest. The reason is structural: high-degree or high-core nodes are not necessarily close, in the effective-resistance sense, to the weak modes that dominate the grounded trace-of-inverse objective. The large improvement over random leaders also shows that leader placement remains important even when the graph is not a low-degree radial-like network.

6.4 Bounded edge-weight reallocation

Table 6 reports bounded edge-weight reallocation for the weighted admittance graphs. The 662-bus and 1138-bus experiments use 20 projected-gradient iterations. The newly added 685-bus row reports a short two-step diagnostic: even this conservative run lowers ten-leader coherence from 1.1493 to 0.9805, a 14.69% additional reduction. Because this row is not a convergence claim, it is interpreted only as evidence that the 685-bus graph also contains useful edge-weight reallocation directions.

Table 6: Bounded edge-weight reallocation after fixing the first ten exact greedy leaders. Pattern graphs are excluded because they do not contain observed admittance-magnitude weights.

Dataset	Edges	Bounds	Initial H	Final H	Reduction	Steps	Lower active	Upper active	Note
662-bus	906	$0.5w_e^{(0)} \le w_e \le 1.5w_e^{(0)}$	2.4992	1.6808	32.75%	20	792	32	20-step PGD
685-bus	1282	$0.5w_e^{(0)} \le w_e \le 1.5w_e^{(0)}$	1.1493	0.9805	14.69%	2	11	249	two-step diagnostic
1138-bus	1458	$0.5w_e^{(0)} \le w_e \le 1.5w_e^{(0)}$	24.7609	16.5082	33.33%	20	1414	0	20-step PGD

The weight experiment is not an engineering line-upgrade prescription. It ignores power-flow

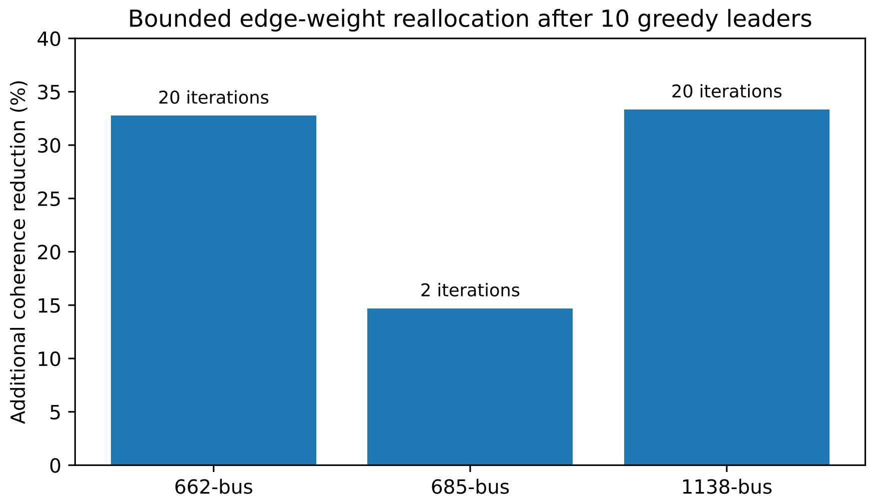


Figure 4: Additional coherence reduction from bounded edge-weight reallocation. The 685-bus bar is a two-step diagnostic rather than a fully converged optimization run.

feasibility, thermal constraints, investment cost, voltage limits, protection coordination, and contingency security. Its role is to quantify the marginal value of coupling weight under the same trace-of-inverse surrogate used for leader placement. Because edges joining two leaders have zero reduced incidence vector, their weights do not directly affect follower covariance; in a physical upgrade study such edges would need separate engineering constraints and cost models.

7 Discussion

7.1 Why centrality is not enough

Degree and weighted degree emphasize local connectivity. Betweenness emphasizes shortest-path brokerage. Harmonic centrality emphasizes distance to all other nodes. These scores are useful diagnostics, but they do not optimize the trace of the inverse grounded Laplacian. Coherence depends on how grounding one node changes all variance modes of the follower system. This makes the objective sensitive to effective-resistance-like structure, bridges, weak corridors, and redundancy among leaders.

The new eris1176-LCC result is the clearest example. Degree-based selection is almost flat between $k = 10$ and $k = 20$, while exact greedy continues to reduce coherence substantially. The reason is that many high-degree nodes lie in already well-connected regions, whereas the best leaders anchor distinct weak modes. This behavior would be hard to diagnose from degree or core number alone.

7.2 Role of the machine-learning screen

The random forest does not replace the spectral objective. It learns a surrogate for exact marginal gains and uses the surrogate only to decide which candidates deserve exact evaluation. This hybrid design is important for trustworthiness: final leader choices are still based on the true Schur-complement gain. The empirical results show that the feature set captures enough information to reduce the candidate pool to roughly one quarter while preserving nearly all of the exact greedy performance.

7.3 Interpretation for power networks

The results should be interpreted as graph-level robustness analysis, not as complete power-system planning. The HB admittance magnitudes provide natural coupling weights, but the model does not include AC power flow, voltage magnitudes, phase angles, generator and load limits, relay protection, N-1 contingency constraints, or economic costs. The pattern graphs contain only topology. Therefore, the selected leaders are best understood as candidate reference or monitoring nodes in a linear stochastic consensus surrogate. Engineering deployment would require additional power-system simulation and feasibility checks.

7.4 Limitations and pre-submission strengthening

Several limitations remain. First, the datasets do not provide operating points, generator/load metadata, bus coordinates, or line limits. Second, the stochastic disturbance is assumed to be identity white noise; heterogeneous or correlated disturbance models may change leader rankings. Third, the eris1176 graph is handled through its largest connected component; a component-aware version of the objective would be more faithful to the full disconnected pattern matrix. Fourth, the 685-bus edge-reallocation result is diagnostic rather than converged. Fifth, the random baseline uses only 12 samples and should be expanded in a final reproducibility package. Sixth, the random-forest screening should be accompanied by a code repository, fixed random seeds, environment files, and raw result tables for formal submission.

Before submission to a network science or control journal, the manuscript should include a reproducibility package containing the parsing scripts, MatrixMarket checksums, random seeds, result-generation scripts, and figure-generation scripts. For a power-systems engineering journal, additional AC power-flow validation, operating scenarios, line-flow constraints, and contingency analyses would be necessary.

8 Conclusion

This paper provides a graph-surrogate validation of grounded-Laplacian coherence optimization on five sparse power-network graph datasets. The newly added 685-bus admittance graph confirms that exact spectral greedy selection remains strong on a more clustered weighted network. The newly added eris1176 pattern graph exposes a disconnected-component issue and, after largest-connected-component handling, provides a demanding unweighted topology where centrality baselines perform especially poorly. Across all datasets, exact greedy improves substantially over random and centrality-based leader sets, while random-forest screening preserves nearly all performance with about one quarter of the exact candidate evaluations. Bounded edge-weight reallocation further shows that leader placement and coupling allocation interact strongly under the trace-of-inverse objective.

With the above qualifications, the resulting manuscript is suitable as a graph-level robustness and network-control study. Its strongest submission fit is network science, complex systems, or control of network systems. A power-systems engineering submission would require an additional layer of physical validation beyond the graph surrogate.

Data and code availability

The numerical experiments use MatrixMarket archives obtained from the Network Repository power-network page: `https://networkrepository.com/power.php`. The specific files are:

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- `power-eris1176.zip`
- `power-US-Grid.zip`

A formal submission should include the parsing scripts, random seeds, generated tables, figure-generation scripts, and a reproduction guide as supplementary material or in a public repository. Redistribution of raw data should follow the policy of the original data repositories.

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A Proof details

A.1 Positive definiteness of the grounded Laplacian

Let G be connected and all edge weights be positive. For any nonempty leader set S, suppose $x^\top L_{FF} x = 0$. Extend x to $\tilde{x} \in \mathbb{R}^N$ by setting leader entries to zero. Then every edge satisfies $\tilde{x}_i = \tilde{x}_j$. Since G is connected, all entries of $\tilde{x}$ are equal. At least one leader entry is zero, so all entries are zero. Hence $x = 0$, proving $L_{FF} \succ 0$.

A.2 Single-leader trace formula

For a connected Laplacian L, the Moore-Penrose inverse satisfies

$$L^+ = \left(L + \frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)^{-1} - \frac{1}{N}\mathbf{1}\mathbf{1}^\top. \tag{17}$$

If node r is grounded, then for $i, j \neq r$,

$$(L_{\bar{r},\bar{r}}^{-1})_{ij} = L_{ij}^+ - L_{ir}^+ - L_{rj}^+ + L_{rr}^+. \tag{18}$$

Taking the trace and using zero row sums of L^+ gives Eq. (10).

A.3 Schur-complement update

Partition $X = L_{FF}^{-1}$ with candidate p first:

$$X = \begin{bmatrix} \alpha & \beta^\top \\ \beta & \Gamma \end{bmatrix}. \tag{19}$$

After grounding p, the inverse is $X_{-p} = \Gamma - \beta\beta^\top/\alpha$. Therefore,

$$\operatorname{tr}(X) - \operatorname{tr}(X_{-p}) = \alpha + \frac{\|\beta\|_2^2}{\alpha} = \frac{\|X_{p,:}\|_2^2}{X_{pp}}. \tag{20}$$

A.4 Gradient of coherence

Let $A(w) = L_{FF}(S, w)$ and $H(w) = (2N)^{-1} \operatorname{tr}(A(w)^{-1})$. Since $\partial A/\partial w_e = b_{e,F} b_{e,F}^\top$ and $dA^{-1} = -A^{-1}(dA)A^{-1}$,

$$\frac{\partial H}{\partial w_e} = -\frac{1}{2N} \operatorname{tr}\left(A^{-1} b_{e,F} b_{e,F}^\top A^{-1}\right) = -\frac{1}{2N} \|A^{-1} b_{e,F}\|_2^2. \tag{21}$$

B Complete twenty-leader greedy sets

Table 7: Exact greedy leader sets for $k = 20$. IDs are one-indexed.

Dataset	Leaders
662-bus	393, 201, 502, 45, 639, 230, 417, 282, 294, 272, 287, 221, 208, 124, 480, 192, 85, 156, 216, 555
685-bus	180, 341, 295, 237, 487, 100, 553, 376, 545, 511, 307, 12, 40, 94, 407, 456, 329, 250, 44, 423
1138-bus	536, 877, 395, 861, 1089, 241, 856, 772, 608, 100, 796, 1013, 590, 601, 558, 862, 857, 218, 615, 337
eris1176-LCC	752, 982, 379, 501, 1127, 1003, 936, 526, 495, 80, 835, 347, 467, 307, 934, 321, 235, 137, 401, 197
US Grid	1243, 4164, 3312, 316, 2554, 726, 4345, 1166, 4652, 1005, 4029, 3785, 2542, 4458, 1653, 2282, 3347, 2586, 657, 803

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